BBC4941 Paper A 2021/22

Question 1 [45 marks]

1. If A, B and C are mutually independent with $P(A)=1/2$, $P(B)=1/3$ and $P(C)=1/4$, $P(AB|C)=$ _____.

$P(A)P(B)P(C)=P(ABC)$

$P(AB|C)=\frac{P(ABC)}{P(C)}$

$A\perp B \Rightarrow A\perp \bar{B},\ \bar{A}\perp B,\ \bar{A}\perp\bar{B}$

2. Let $A_n=\left\{X\in\left(1-\frac{1}{n},1\right)\right\}, n=1,2,3,\cdots$, where X is normally distributed with mean 0 and variance 4. Then $P(\bigcap_{n=1}^{\infty}A_n)=$ _____.

$A_n \downarrow \Phi$ $\quad \lim_{n\to\infty} P(A_n)=0$

$E=\frac{1}{\beta}$

$Exp(\beta)$

3. Let random variable X be geometrically distributed, that is $P(X=k)=p(1-p)^{k-1}, k=1,2,\cdots$. Then $P(X>m+n\,|\,X>m)=$ _____.

$P(X>n)=\sum_{k=n+1}^{\infty}p(1-p)^{k-1}=p\frac{(1-p)^n}{(1-(1-p))}$

4. Suppose that $X_1,X_2,\cdots,X_n$ form a sequence of Bernoulli random variables with parameter p such that $P(X_k=1)=p, P(X_k=0)=1-p, k=1,2,\cdots$ and $Y=\sum_{k=1}^{n}X_k$. As we know that Y obey binomial distribution. Then $P(X_n=1|Y=1)=$ _____.

$Y\sim B(n,p)$

$\frac{P(X_n=1, Y=1)}{P(Y=1)}=\frac{P(X_1=0,\cdots X_{n-1}=0, X_n=1)}{C_n^1 p q^{n-1}}$

5. Suppose that random variables X and Y are independent and both obey the standard normal distribution ($\Phi(x)$ is the distribution function). Then $P(X+Y\le 1)$ is _____.

$X+Y\sim N(0,2)$

$N(\mu_1,\mu_2,\sigma_1,\sigma_2)$

$N(\mu,\sigma^2)$

$=P\left(\frac{X+Y-0}{\sqrt{2}}\le\frac{1-0}{\sqrt{2}}\right)$

$=P\left(\le\frac{1}{\sqrt{2}}\right)=\Phi\left(\frac{1}{\sqrt{2}}\right)$

6. Suppose that X has an exponential distribution with mean 1. The p.d.f. of $Y=\ln(X+1)$ is _____.

$e^{y}-1=x=h(y)$

(ii) $f_Y(y)=\begin{cases}e^{-(e^y-1)}e^y, & y>0\\ 0, & \text{otherwise}\end{cases}$

7. Suppose that the distribution function (d. f.) of random variables X is as follows:

$$F(x)=\begin{cases}0, & x<-\frac{\pi}{2},\\ 1/8, & x=-\frac{\pi}{2},\\ a\sin x+b, & -\frac{\pi}{2}<x<\frac{\pi}{2},\\ 1, & x\ge\frac{\pi}{2}.\end{cases}$$

$F(\frac{\pi}{2}-0)=\frac{1}{4}$

$F(-\frac{\pi}{2})=\frac{1}{8}$

$P(X<\frac{\pi}{2})=1/4$. Then $a=$ _____.

$\Phi(x)+\Phi(-x)=1$

$\Phi(0)=\frac{1}{2}$

$2X+5Y-3\sim N(E$, Var)

all the properties

$X+Y,\ X\cap Y,\ X\cup Y$

p.d.f.; d.f., p.l.

p.d.f

$\int f=1$

8. Suppose that the p.d.f. of random variable X is defined as follows:

$$f(x)=\begin{cases}1/2, & 0\le x\le 1,\\ 1/6, & 3\le x\le 6,\\ 0, & \text{otherwise.}\end{cases}$$

If $P(X\ge a)=1/2$, please give a feasible range of a: $[1,3]$

E, Var, (Defs, properties)

Cov(X, Y)

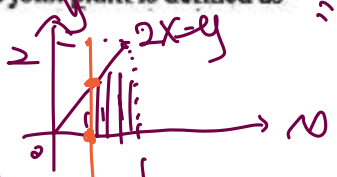
ρ_{XY}

9. With the same p.d.f. as in above Problem 8, $E[e^X]=$ _____.

$\int_{-\infty}^{\infty}e^x f(x)dx=\int_0^1\frac{1}{2}e^x dx+\int_3^6\frac{1}{6}e^x dx$

10. Suppose that X and Y have a continuous joint distribution for which the joint p.d.f. is defined as follows:

$$f(x,y)=\begin{cases}\frac{1}{3}(x+y), & 0<x<1, 0<y<2,\\ 0, & \text{otherwise.}\end{cases}$$



Then $P(2X>Y)=$ _____.

A

$\int_{\{2x>y\}}f(x,y)dxdy=\int_0^1 dx\int_0^{2x}\frac{1}{3}(x+y)dy$

11. Under the same conditions as in Problem 10 above, is X independent of Y? (answer Yes or No) _____.

$\forall(x,y)\ F(x,y)=F_X(x)\cdot F_Y(y)$;

$f(x,y)=f_X(x)\cdot f_Y(y)$

(x,y). x, y conti.

12. Under the same conditions as in Problem 10 above, $Cov(X,Y)=$ _____.

$E(XY)-E(X)E(Y)$

$=\int_{\mathbb{R}^2}xy\,f(x,y)dxdy-\int_{\mathbb{R}^2}x\,f(x,y)dxdy\cdot$ _____

$\rho_{XY}=\rho_X\cdot\rho_Y$

CLT: $\sum_{i=1}^{n}X_i\sim N(\quad,\quad)$ approximately

$N(t)-N(s) \sim P(\lambda(t-s))$ $\quad N(t) \sim P(\lambda t)$

$$\boxed{C_N(s, t+s) = Var_N(s \wedge t+s) = Var_N(s) = \lambda s}$$

$N(s)$

13. Suppose that $\{N(t), t \geq 0\}$ is a Poisson process with parameter $\lambda = 1$. Then the autocorrelation function $R_N(s, t+s) = C_N(s, t+s) + \mu_N(s)\cdot\mu_N(t+s) = \lambda s + \lambda s \cdot \lambda(t+s)$

14. Consider a Markov chain $\{X_n, n = 0,1,2,\cdots\}$ with the state space $\{1,2,3\}$, initial distribution $\{X_0, X_1, \cdots\}$, $\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$ and the matrix of transition probabilities $P = \begin{bmatrix} \frac{3}{4} & 1 & 0 \\ \frac{1}{4} & 1 & \frac{1}{4} \\ 0 & 1 & \frac{1}{4} \end{bmatrix}$. Then $P(X_1 > 1) =$ $P(X_1=2) + P(X_1=3)$

P^2 $\quad P(X_2>1) =$

$P(X_n = i) = \sum_{j=1}^{3} P_j^{(0)} P_{ji}^{(n)}$ $\quad P(X_2=2) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}) \begin{pmatrix} \frac{1}{4} \\ 1 \\ \frac{1}{4} \end{pmatrix}$

15. Suppose that the process $\{X(t), -\infty < t < +\infty\}$ is a stationary process with the autocorrelation function $R_X(t) = ae^{-b|t|}$, where a, b are positive constants. Then the average power $Q =$ _____

$Q = \psi_X^2(0) = R_X(0)$ $\quad P(X_2=2) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}) \begin{pmatrix} \frac{1}{4} \\ 1 \\ \frac{1}{4} \end{pmatrix}$

stationary distribution

$\pi = \pi P$

$\pi_1 + \pi_2 + \pi_3 = 1$

$\lim_{n\to\infty} P(X_n = 2) = \pi_2$

$\lim_{n\to\infty} P(X_n) = \pi_2$

$= P(X_\infty = 2)$

Question 2. [11 marks]

$\{X_t\}$ Random variable (X, Y) has a joint p.d.f as follows:

first-order distribution

$f(x; t) =$

$F(x; t)$

$$f(x,y) = \begin{cases} \frac{21}{4}x^2 y, & x^2 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal p.d.f of X; calculate $P\{Y > 3/4 \mid X = 1/2\}$.

$f_X(x)$, $f_Y(y)$; uncorrelated? independent?

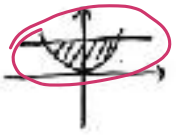
$Cov(X,Y) = 0$

Question 2.

k? $(\int f = 1)$

x: $F_X(x)$?

$P(X \in A) = ?$ $\quad P(X, Y \in A)$



(1) $f_X(x) = \int_{x^2}^{1} \frac{21}{4} x^2 y \, dy = \frac{21}{8} x^2 (1 - x^4)$

$$f_X(x) = \begin{cases} \frac{21}{8} x^2 (1-x^4) & -1 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

discrete r.v. $p.f.$ $X^2 + Y$; E, Var, ρ_{XY}, $Cov(X,Y)$

(2) $P(Y > \frac{3}{4} \mid X = \frac{1}{2})$.

$\boxed{\text{if } x \in}$

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{21}{4}x^2 y}{\frac{21}{8}x^2(1-x^4)} = \frac{2y}{1-x^4}$$

$Y \mid X$

$$f_{Y|X}(y|x) = \begin{cases} \frac{2y}{1-x^4} & x^2 < y < 1, \ -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Let $x = \frac{1}{2}$

$f_{Y|\frac{1}{2}}(y|\frac{1}{2}) = \frac{2y}{1-\frac{1}{16}} = \frac{32}{15}y = \{$

$P(Y > \frac{3}{4} \mid x = \frac{1}{2}) = \int_{\frac{3}{4}}^{1} \frac{32}{15} y \, dy = \frac{7}{15}$

$\int_{\frac{1}{4}}^{\infty} f_{Y|X}(y|\frac{1}{2}) \, dy$

Question 3. [11 marks]

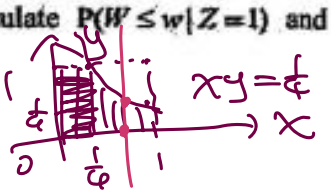
X and Y are two numbers randomly and independently chosen from $[0,1]$. The p.d.f of X is as follows:

$$f(x)=\begin{cases}1, & 0\le x\le 1,\\ 0, & \text{otherwise.}\end{cases}$$

$$f(x,y)=\begin{cases}1, & 0\le x,y\le 1\\ 0, & \text{otherwise}\end{cases}$$

Find the probability $P(XY<1/4)$ by the joint p.d.f. of (X,Y). Define random variable $Z(\omega)=\begin{cases}1, & X(\omega)\le 1/2,\\ 0, & \text{otherwise.}\end{cases}$ Calculate the p.f. of Z. Let $W=XZ$. Calculate $P(W\le w\mid Z=1)$ and $P(W\le w\mid Z=0)$ for $0\le w\le \frac{1}{2}$. Then find d.f. of W.

p.1.



(1) $P(XY<\frac{1}{4})$

$$=\int_{xy<\frac{1}{4}} f(x,y)\,dx\,dy = 1\times\frac{1}{4}+\int_{\frac{1}{4}}^{1}dx\int_0^{\frac{1}{4}\frac{1}{x}}dy$$

(2) $P(Z=1)=P(X\le\frac{1}{2})=\frac{1}{2}$

$P(Z=0)=1-P(Z=1)=\frac{1}{2}$

(3) $P(W\le w\mid Z=1)=P(X\le w\mid X\le\frac{1}{2})=$ _____

(4) $P(W\le w)=P(W\le w\mid Z=1)P(Z=1)$

$+P(\quad\mid Z=0)P(Z=0)$

Question 4. [11 marks] Suppose that $\{X_n, n \geq 0\}$ is a discrete time and homogeneous Markov chain with state space $S = \{1,2,3\}$, and the following transition probability matrix:

$$P = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \end{pmatrix} \end{array}.$$

(π_1, π_2, π_3)

Is $\{X_n, n \geq 0\}$ ergodic? Why? Find the stationary distribution (if it exists) and the probability Calculate $P\{X_3 = 1, X_0 = 2 | X_2 = 3\}$ with initial distribution: $P\{X_0 = 1\} = P\{X_0 = 2\} = P\{X_0 = 3\} = 1/3$.

Question 4

(1) $P^2 = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} = \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{3}{4} & 0 & \frac{1}{4} \end{pmatrix} \end{array}$

$P^4 = P^2 \cdot P^2$

$P^3 = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{3}{4} & 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} \frac{5}{8} & \frac{1}{8} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{8} & \frac{3}{8} \end{pmatrix}$

$\therefore$ There is no 0 in $P(3)$. It is ergodic.

(2) Let be $\pi = (\pi_1, \pi_2, \pi_3)$ $\quad \pi = \pi P$, $\pi_1 + \pi_2 + \pi_3 = 1$

(3) $P(X_3 = 1, X_0 = 2 | X_2 = 3)$

$= \dfrac{P(X_3 = 1, X_0 = 2, X_2 = 3)}{P(X_2 = 3)}$ $\quad P(X_0 = 2, X_2 = 3, X_3 = 1)$

$P_2(0) \; P_{23}(2) \; P_{31}$

$= \dfrac{P(X_0 = 2) \cdot P_{23}(2) \cdot P_{31}(1)}{P_1(0) P_{13}(2) + P_2(0) P_{23}(2) + P_3(0) P_{33}(2)}$

$= \dfrac{\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{2}}{\frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{3}{8}} = \dfrac{\frac{1}{24}}{\frac{16}{48}} = \dfrac{1}{7}.$

$P(X_2 = 3) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{4} \end{pmatrix}$

$E(X_t \cdot X_{t+\tau})$

$R_X(\tau) \longleftrightarrow S_X(\omega)$

$S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-i\omega\tau} d\tau \Leftrightarrow R_X(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega) e^{i\omega\tau} d\omega$

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Question 5. [11 marks] Suppose that process $\{X(t), t \geq 0\}$ is wide-sense stationary with autocorrelation function $R_X(t) = e^{-|t|}$. Find the power spectral density of $\{X(t), t \geq 0\}$. Let $Y(t) = X(t+2) - X(t)$. Prove that $\{Y(t), t \geq 0\}$ is a wide-sense stationary process.

Question 5.

(1) $S_X(\omega) = \int_{-\infty}^{\infty} R_X(\tau) e^{-i\omega\tau} d\tau$
$= \frac{2}{1+\omega^2}$

$R_X(\tau) \longleftrightarrow S_X(\omega)$

$e^{-a|\tau|} \longleftrightarrow \frac{2a}{a^2+\omega^2}$

$1 \longleftrightarrow 2\pi\delta(\omega)$

$\delta(\tau) \longleftrightarrow 1$

$\cos(\omega_0\tau) \Leftrightarrow \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$

(1) $Q = R_X(0) = 1$

$g(X)$

(2) $E(Y(t)) = E(X(t+2)) - E(X(t)) = \text{Constant}$

$\because X(t)$ is wide-sense stationary.

$\therefore E(X(t)) = \mu_X$.

$E(Y(t)) = 0$.

$\tilde{g}(X)$

$R_Y(t, t+\tau) = E(Y_t \cdot Y_{t+\tau})$

$= E[(X(t+2) - X(t)) \cdot (X(t+2+\tau) - X(t+\tau))]$

$= E[X(t+2)X(t+2+\tau) - X(t+2)X(t+\tau) - X(t)X(t+2+\tau) + X(t)X(t+\tau)]$

$= R_X(\tau) - 2R_X(2+\tau) - R_X(\tau)$

$= -R_X(2+\tau) + 2R_X(\tau) - R_X(\tau-2) =: R_Y(\tau)$

$\therefore Y(t)$ is a wide-sense stationary process

$Z(0) = X \sim N(\mu, \sigma^2)$

$Z(\frac{1}{4}) = Y \sim N(\mu, \sigma^2)$

Question 6 [11 marks] Let $Z(t) = X\cos 2\pi t + Y \sin 2\pi t$, where X, Y are independent variables and obey the same normal distribution with mean μ and variance σ^2. Actually, $Z(t)$ is a Gaussian process. Find the $\boxed{\text{p.d.f of } Z(t)}$ and the joint p.d.f of $(Z(0), Z(1/4))$. Calculate $Var(Z(t))$ and $Cov(Z(t), Z(s))$.

$Z(t) \sim N(\mu_2, \sigma^2)$

$f_Z(x;t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_2)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$

Question 6

$Z(0) \sim N(\mu, \sigma^2)$

$Z(\frac{1}{4}) \sim N(\mu, \sigma^2)$

$f(Z(0), Z(\frac{1}{4}))$

$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$\cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$= \frac{1}{2\pi\sigma^2} e^{-\frac{(x-\mu)^2 - (x-\mu)^2}{2\sigma^2}}$

$X \sim (\mu, \sigma^2) \qquad Y \sim (\mu, \sigma^2) \qquad Z(t) \sim N(\mu(\cos 2\pi t + \sin 2\pi t), \sigma^2)$

$E(Z(t)) = E(X)\cos 2\pi t + E(Y)\sin 2\pi t = \mu(\cos 2\pi t + \sin 2\pi t)$

$E(Z(t)Z(t)) = E(X^2)\cos^2 2\pi t + E(Y^2)\sin^2 2\pi t$

$E(X^2) = Var(X) + E^2(X) = \sigma^2 + \mu^2 = E(Y^2)$

$E(Z^2(t)) = \sigma^2 + \mu^2$

$Var(Z(t)) = Var(X)\cos^2 2\pi t + Var(Y)\sin^2 2\pi t = \sigma^2$

$Cov(Z(t), Z(s)) = E[(Z(t) - E(Z(t)))(Z(s) - E(Z(s)))]$

$= E[(Z(t) - \mu(\cos 2\pi t + \sin 2\pi t))(Z(s) - \mu(\cos 2\pi s + \sin$

$= C_X(Z(t), Z(s)) = R_X(Z(t), Z(s)) - E(Z(t))E(Z$

$R_X(Z(t)Z(s)) = E(X^2)\cos 2\pi t \cos 2\pi s + E(Y^2)\sin 2\pi t \sin 2\pi s.$

$= (\sigma^2 + \mu^2)\cos(2\pi(t-s))$

$Cov(Z(t), Z(s)) = (\sigma^2 + \mu^2)\cos[2\pi(t-s)] - \mu^2(\cos 2\pi(t-s))$

$= \sigma^2 \cos[2\pi(t-s)]$